Q. Implement following:

A. WAP to print twin primes less than 1000. If two consecutive odd numbers are both prime then they are known as twin prime.

B. Write a program to implement these formulae of permutations and combinations. Number of permutations of n objects taken r at a time: p(n, r) = n! / (n-r)!. Number of combinations of n objects taken r at a time is: c(n, r) = n! / (r!*(n-r)!) = p(n,r) / r!

1. **Code for experiment/practical:**

```
2.  Code to Calculate twin prime number.
3.  for num in range(3, 1000):
4.  # Check if 'num' is prime
5.  isPrime1 = True 6. for i in range(2, num):
7.  if num % i == 0:
8.  isPrime1 = False
9.  break
10.
11. # Check if 'num + 2' is prime
12. isPrime2 = True
13. for i in range(2, num + 2):
14. if (num + 2) % i == 0:
15. isPrime2 = False
16. break
```

```
Code to Calculate Permutations and Combinations

# Input the values of n and r
n = int(input("Enter value of n: ")) r = int(input("Enter value of r: "))

# Calculate factorial of a number
n_fact = 1
for i in range(1, n + 1): n_fact *= i

n_r_fact = 1
for i in range(1, n - r + 1): n_r_fact *= i

r_fact = 1
for i in range(1, r + 1): r_fact *= i

# Permutation formula: p(n, r) = n! / (n-r)!
permutations = n_fact // n_r_fact

# Combination formula: c(n, r) = n! / (r! * (n-r)!)
combinations = permutations // r_fact

# Output the results
print(f"Permutations p({n}, {r}) = {permutations}") print(f"Combinations
    c({n}, {r}) = {combinations}")
```

2. **Result/Output/Writing Summary:**

Output for Twin prime number

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

The twin Prime Numbers are =  881 883
PS E:\Worksheets\Python> python -u "e:\Worksheets\Python\twinPrime.py"
The twin Prime Numbers are -> 3 5
The twin Prime Numbers are -> 5 7
The twin Prime Numbers are -> 11 13
The twin Prime Numbers are -> 17 19
The twin Prime Numbers are -> 29 31
The twin Prime Numbers are -> 41 43
The twin Prime Numbers are -> 59 61
The twin Prime Numbers are -> 71 73
The twin Prime Numbers are -> 101 103
The twin Prime Numbers are -> 107 109
The twin Prime Numbers are -> 137 139
The twin Prime Numbers are -> 149 151
The twin Prime Numbers are -> 179 181
The twin Prime Numbers are -> 191 193
The twin Prime Numbers are -> 197 199
The twin Prime Numbers are -> 227 229
The twin Prime Numbers are -> 239 241
The twin Prime Numbers are -> 269 271
The twin Prime Numbers are -> 281 283
The twin Prime Numbers are -> 311 313
The twin Prime Numbers are -> 347 349
The twin Prime Numbers are -> 419 421
The twin Prime Numbers are -> 431 433
The twin Prime Numbers are -> 461 463
The twin Prime Numbers are -> 521 523
The twin Prime Numbers are -> 569 571
The twin Prime Numbers are -> 599 601
The twin Prime Numbers are -> 617 619
The twin Prime Numbers are -> 641 643
The twin Prime Numbers are -> 659 661
The twin Prime Numbers are -> 809 811
The twin Prime Numbers are -> 821 823
The twin Prime Numbers are -> 827 829
The twin Prime Numbers are -> 857 859
The twin Prime Numbers are -> 881 883
PS E:\Worksheets\Python>
```

Output for Combination and Permutation

```
PS D:\Worksheets> python -u "d:\Worksheets\Python\permutation.py"
Enter value of n: 5
Enter value of r: 3
Permutations p(5, 3) = 60
Combinations c(5, 3) = 10
PS D:\Worksheets>
```

3. **Learning outcomes (What I have learnt):**
    1. Learned about Twin prime number.
    2. Learned about how range function works.
    3. Learned about for loop.